SIT313 – Task P4: Login and Registration Page

Student Name: Lav
Unit: SIT313 – Secure Frontend Applications
Task: P4 – Login and Registration Page

1. Introduction

The goal of this task was to create a login and registration page, for the DEV@Deakin website using React and Firebase. This task helped me learn how people can sign up for an account log in using their existing information and keep user details stored in a database.

2. Technologies Used

- **React:** Used to create the website components and login and registration forms.
- **React Router:** Used to navigate between the homepage, login page and signup page.
- **Firebase Authentication:** Used to create accounts and verify login details.
- **Cloud Firestore:** Used to store registered users' names and email addresses.
- **CSS:** Used to style the navigation bar and authentication forms.
- **GitHub:** Used to store and manage the project code.

3. Implementation

3.1 Navigation Bar

I modified the navigation bar to display the DEV@Deakin link, a search field, a Post button and a Login link. The DEV@Deakin link takes users back to the homepage. The Search and Post features are currently visual elements because their functionality will be implemented in a future task.

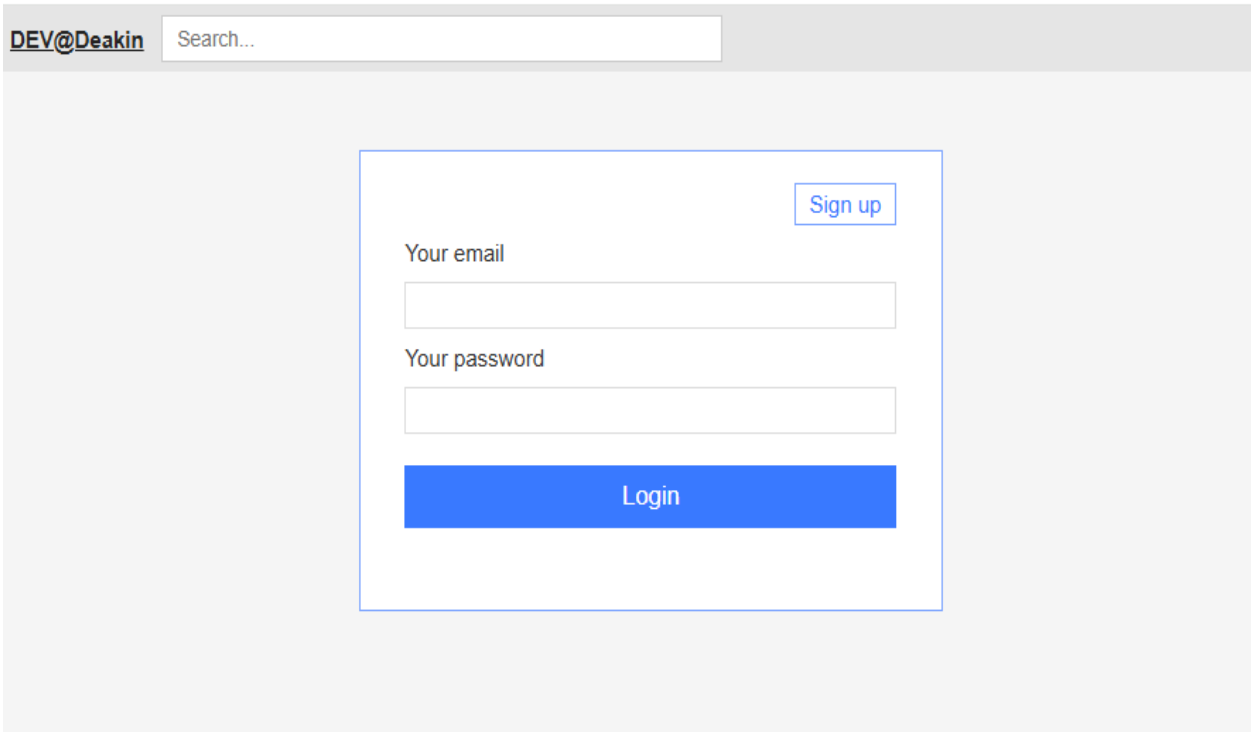
Screenshot 1: Navigation Bar



3.2 Login Page

I created a login form that allows existing users to enter their email and password. Firebase authentication checks the login details. When the login is successful, the user is redirected to the homepage. If the details are incorrect, an error message asks the user to check the details.

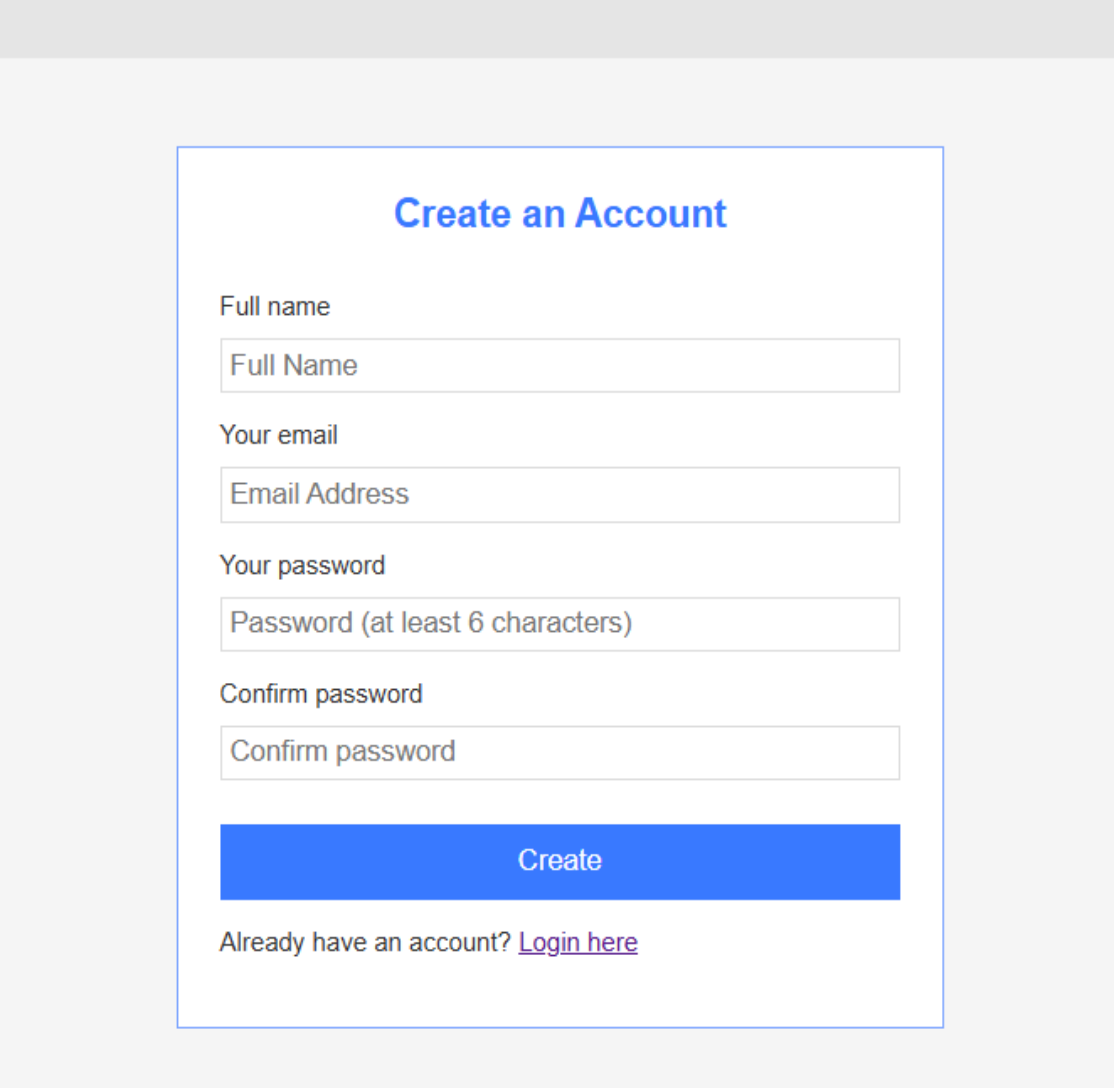
Screenshot 2: Login Form



3.3 Registration Page

I created a registration form that collects the user's name, email, password and confirmation of password. The form checks the password and confirmation match before accepting registration.. Firebase authentication creates the account, and the user's name and email address are saved in Cloud Firestore. Ater successful registration , the user is redirected to the login page..

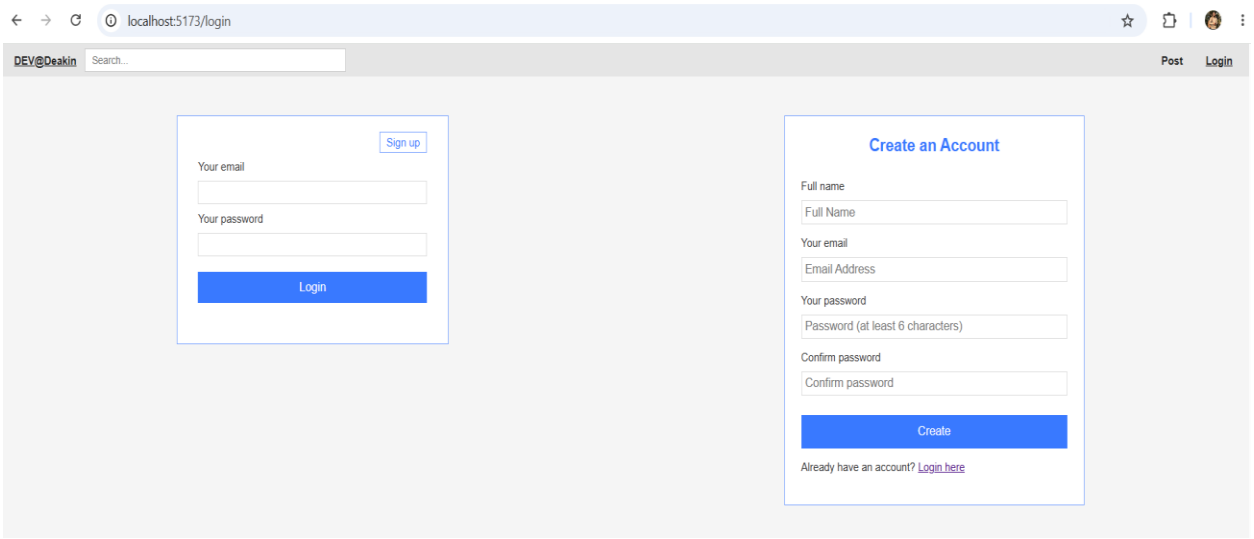
Screenshot 3: Registration Form



3.4 Navigation and Page Layout

I used React components to keep the navigation bar and footer consistent across the pages. React Router manages the homepage, login and signup routes. I also adjusted the CSS to display the login and registration forms side by side.

Screenshot 4: Combined Login and Registration Layout



4. Testing

I tested the main functions of the website to check that they worked as expected.

Test	Expected Result	Result
Open the login page	Login form appears	Passed
Enter correct login details	User is redirected to the homepage	Passed
Enter incorrect login details	An error message appears	Passed
Register with different passwords	Password mismatch message appears	Passed
Register with matching passwords and a new email	Account is created successfully	Passed
Check Firestore user details	User's name and email are saved	Passed
Open the login route	Both authentication panels appear	Passed

5. GitHub Repository

The project code is available in my GitHub repository.

Repository link:
https://github.com/lavsukhija001-sudo/SIT313-Task-P4

6. Conclusion

In conclusion, I developed and tested the main login and registration features for the DEV@Deakin website. The application supports account registration, login validation and storage of user profile details. I also updated the navigation bar and page layout and prepared the project for submission through GitHub.